SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A

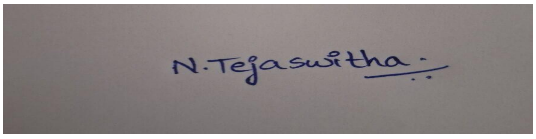
Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools ***(2)***	Analysis of Results ***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _________N.Tejaswitha 192371057___________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

	Photograph:		
Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17

Player ID	Player Name	Points Scored
P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

QUESTION 01: Photograph Preference Analysis

1. Introduction

The given data represents the preferences of children from three different age groups for three photographs A, B and C. R programming is used to calculate the sample covariance and correlation between preferences B and C, along with their covariance and correlation matrices. A Chi-square test is also performed to determine whether there is a significant relationship between age group and photograph preference.

2. R Program

```
# Enter data

data <- data.frame(
Age = c("5-6", "7-8", "9-10"),
A = c(18, 2, 20),
B = c(22, 28, 10),
C = c(20, 40, 40)
)

# Display data

print(data)

# 1. Covariance between B and C

cov_B_C <- cov(data$B, data$C)
print(cov_B_C)

# 2. Covariance matrix

cov_matrix <- cov(data[, c("A", "B", "C")])
print(cov_matrix)

# 3. Correlation between B and C

cor_B_C <- cor(data$B, data$C)
print(cor_B_C)
```

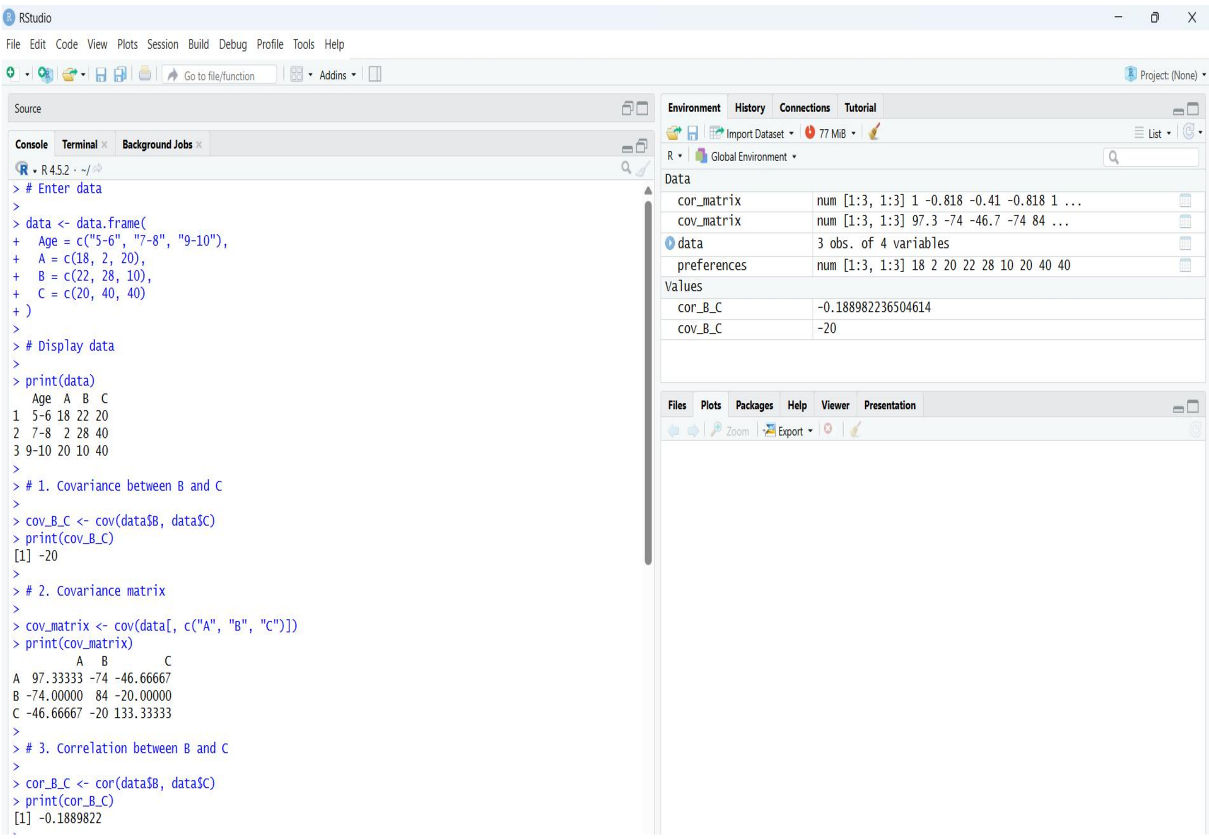
 # 4. Correlation matrix

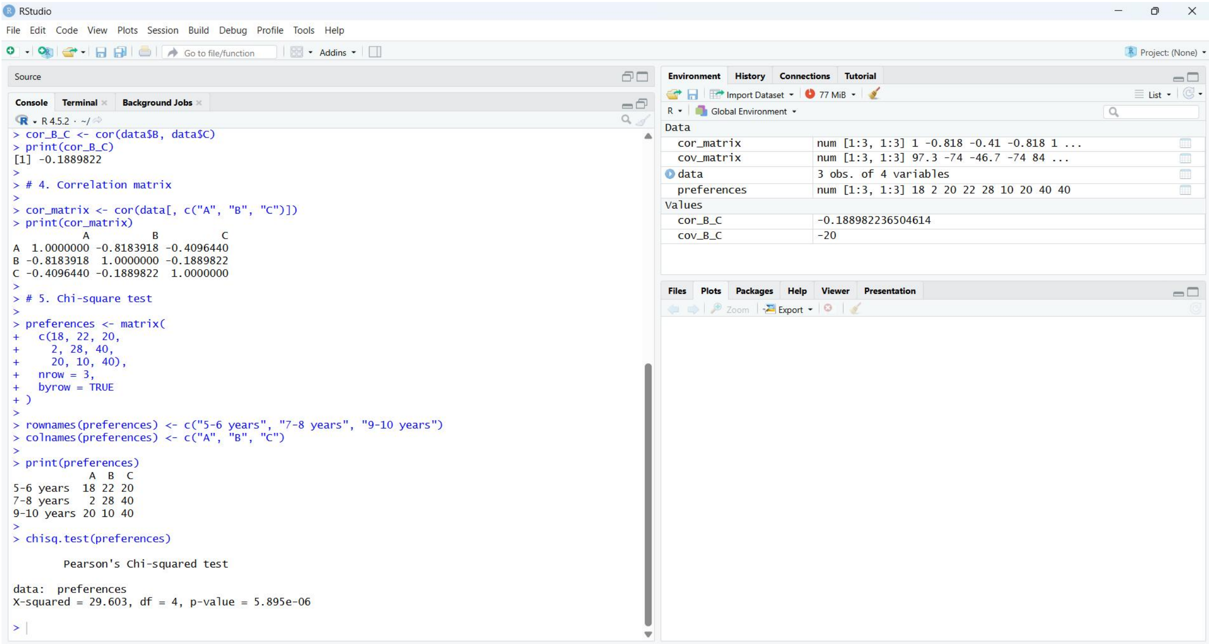
```
cor_matrix <- cor(data[, c("A", "B", "C")])
print(cor_matrix)
```

 # 5. Chi-square test

```
preferences <- matrix(
c(18, 22, 20,
2, 28, 40,
20, 10, 40),
nrow = 3,
byrow = TRUE
)

rownames(preferences) <- c("5-6 years", "7-8 years", "9-10 years")
colnames(preferences) <- c("A", "B", "C")

print(preferences)

chisq.test(preferences)
```

3. OUTPUT





4. Results and Interpretation

- ✧ The **sample covariance between B and C is -20**. The negative value indicates that the preferences for photographs B and C vary in opposite directions in the given data.
- ✧ The **sample correlation between B and C is approximately -0.189**, indicating a weak negative relationship between the two preferences.
- ✧ The covariance and correlation matrices show the relationships among preferences A, B and C.
- ✧ The Chi-square test gives a **p-value of 5.895×10^{-6}**, which is less than 0.05, indicating a statistically significant relationship between age group and photograph preference.

5. Limitations of the Study

The study may be affected by other factors such as the **attractiveness, familiarity, appearance, and presentation order** of the photographs. The method of selecting the children and controlling these factors is also not clearly specified. Therefore, age alone cannot necessarily be considered the cause of the difference in photograph preference.

6. Conclusion

The analysis shows a **significant relationship between children's age group and photograph preference.** The covariance and correlation calculations describe the relationship between the photograph preferences. However, the study has limitations because other factors influencing children's choices may not have been controlled.

QUESTION 02: Tennis Players' Points and Outlier Analysis

1. Introduction

The given data represents the points scored by 15 tennis players. R programming is used to visualize the distribution of points using a boxplot and identify whether any player has scored an outlier.

2. R Program

```
# Enter data
data <- data.frame(Player_ID = c("P1", "P2", "P3", "P4", "P5",
        "P6", "P7", "P8", "P9", "P10",
        "P11", "P12", "P13", "P14", "P15"),Player_Name = c("Player A", "Player B",
"Player C", "Player D", "Player E",
         "Player F", "Player G", "Player H", "Player I", "Player J",
         "Player K", "Player L", "Player M", "Player N", "Player O"),Points_Scored
= c(12, 15, 14, 10, 18,
          20, 22, 13, 11, 35,
          16, 17, 19, 21, 23)
)

# Display data
print(data)
# 1. Boxplot of points scored
boxplot(data$Points_Scored,
    main = "Boxplot of Points Scored",
    ylab = "Points Scored",
    col = "lightblue")
# 2. Identify outliers
outliers <- boxplot.stats(data$Points_Scored)$outprint(outliers)
# 3. Five-number summary
summary(data$Points_Scored)
```

3. OUTPUT

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

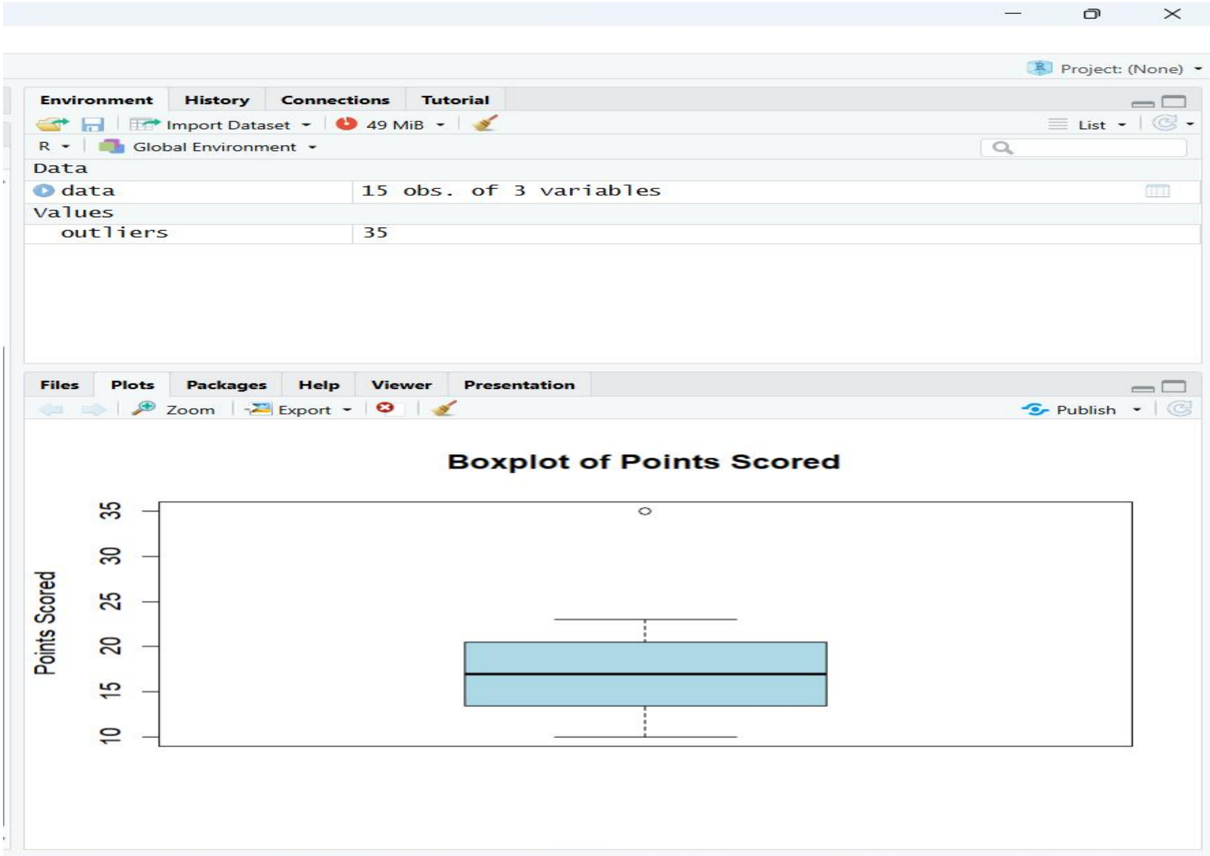
Source

Console Terminal Background Jobs

R 4.5.2 · ~/

```
> # Enter data
>
> data <- data.frame(
+   Player_ID = c("P1", "P2", "P3", "P4", "P5",
+                 "P6", "P7", "P8", "P9", "P10",
+                 "P11", "P12", "P13", "P14", "P15"),
+   Player_Name = c("Player A", "Player B", "Player C", "Player D", "Player E",
+                   "Player F", "Player G", "Player H", "Player I", "Player J",
+                   "Player K", "Player L", "Player M", "Player N", "Player O"),
+   Points_Scored = c(12, 15, 14, 10, 18,
+                     20, 22, 13, 11, 35,
+                     16, 17, 19, 21, 23)
+ )
>
> # Display data
>
> print(data)
   Player_ID Player_Name Points_Scored
1         P1    Player A            12
2         P2    Player B            15
3         P3    Player C            14
4         P4    Player D            10
5         P5    Player E            18
6         P6    Player F            20
7         P7    Player G            22
8         P8    Player H            13
9         P9    Player I            11
10       P10    Player J            35
11       P11    Player K            16
12       P12    Player L            17
13       P13    Player M            19
14       P14    Player N            21
15       P15    Player O            23
```

```
> # Display data
>
> print(data)
   Player_ID Player_Name Points_Scored
1         P1    Player A            12
2         P2    Player B            15
3         P3    Player C            14
4         P4    Player D            10
5         P5    Player E            18
6         P6    Player F            20
7         P7    Player G            22
8         P8    Player H            13
9         P9    Player I            11
10       P10    Player J            35
11       P11    Player K            16
12       P12    Player L            17
13       P13    Player M            19
14       P14    Player N            21
15       P15    Player O            23
>
> # 1. Boxplot of points scored
>
> boxplot(data$Points_Scored,
+         main = "Boxplot of Points Scored",
+         ylab = "Points Scored",
+         col = "lightblue")
>
> # 2. Identify outliers
>
> outliers <- boxplot.stats(data$Points_Scored)$out
> print(outliers)
[1] 35
>
> # 3. Five-number summary
>
> summary(data$Points_Scored)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  10.00   13.50   17.00   17.73   20.50   35.00
>
```



4. Results and Interpretation

- ➢ The boxplot represents the distribution of points scored by the tennis players. The value **35** appears as an outlier because it lies beyond the upper whisker of the boxplot. This indicates that **Player J**, who scored 35 points, has an unusually high score compared with the other players.
- ➢ Most players' scores are concentrated between approximately **10 and 23 points**, while the score of 35 is considerably higher than the rest of the observations.

5. Outlier Analysis

The boxplot identifies **35 points** as an outlier. This corresponds to **Player J (P10)**. The unusually high score should be examined to determine whether it represents exceptional performance or whether there may be an error or unusual circumstance in the data.

6. Conclusion

The boxplot successfully visualizes the distribution of points scored by the tennis players and identifies an outlier. **Player J (P10), with 35 points, is the outlier.** Therefore, the coach can investigate this score separately when analyzing the team's overall performance.

QUESTION 03: Decision Tree Classifier for Loan Approval

1. Introduction

A bank wants to predict whether a customer's loan should be approved or rejected based on attributes such as **Age, Income, Employment Status, and Credit Score**. A Decision Tree classifier is used to learn patterns from the historical data and predict **Loan Approval (Yes/No)**. R programming is used to build and visualize the Decision Tree model.

2. R Program

 # Enter data

```
data <- data.frame(

  Age = c("young", "young", "middle", "old", "old",

          "old", "middle", "young", "young", "old",

          "middle", "middle", "old", "young", "middle"),

  Income = c("high", "high", "high", "medium", "low",

             "low", "low", "medium", "low", "medium",

             "medium", "high", "medium", "medium", "low"),

  Employment_Status = c("employed", "self-employed", "employed",

                        "employed", "unemployed", "self-employed",

                        "unemployed", "employed", "unemployed",

                        "self-employed", "employed", "self-employed",

                        "unemployed", "self-employed", "employed"),

  Credit_Score = c("good", "average", "good", "good", "poor",

                   "average", "poor", "average", "poor", "good",

                   "average", "good", "poor", "average", "poor"),

  Loan_Approval = c("yes", "yes", "yes", "yes", "no",
```

```r
                    "no", "no", "yes", "no", "yes",
                    "yes", "yes", "no", "yes", "no")
)
# Display data
print(data)

# Convert variables to factors
data$Age <- as.factor(data$Age)
data$Income <- as.factor(data$Income)
data$Employment_Status <- as.factor(data$Employment_Status)
data$Credit_Score <- as.factor(data$Credit_Score)
data$Loan_Approval <- as.factor(data$Loan_Approval)

# 1. Build Decision Tree classifier
library(rpart)
model <- rpart(
  Loan_Approval ~ Age + Income + Employment_Status + Credit_Score,
  data = data,
  method = "class",
  control = rpart.control(
    cp = 0,
    minsplit = 2,
    minbucket = 1,
    maxdepth = 5
```

```
  )
)
```

 # 2. Display Decision Tree model

```
print(model)
```

 # 3. Display Decision Tree clearly

```
par(mar = c(2, 2, 3, 2))
plot(model,
     uniform = TRUE,
     margin = 0.1)
text(model,
     use.n = TRUE,
     all = TRUE,
     cex = 0.9)
```

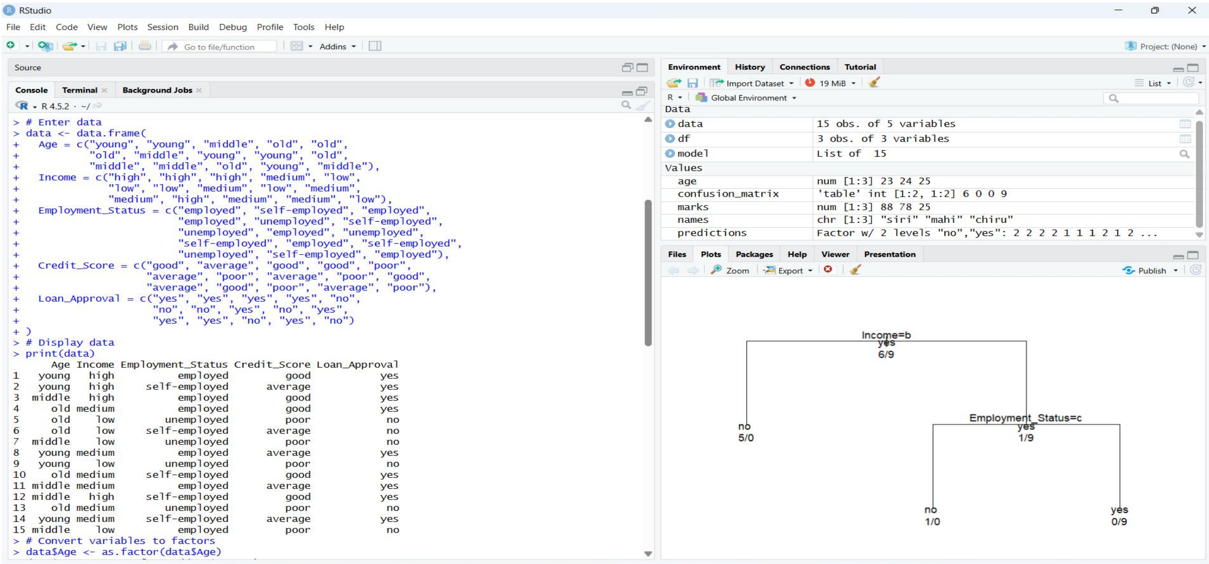
 # 4. Predict Loan Approval

```
predictions <- predict(model, data, type = "class")
print(predictions)
```

 # 5. Confusion Matrix

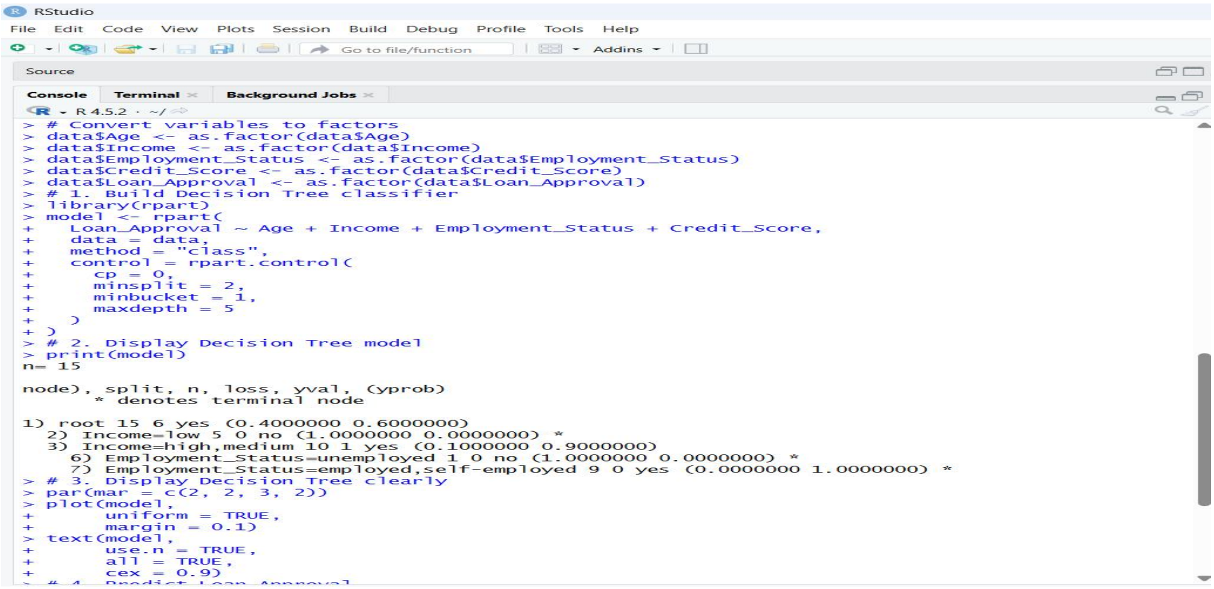
```
confusion_matrix <- table(
  Actual = data$Loan_Approval,
  Predicted = predictions
)
print(confusion_matrix)
```

3. OUTPUT

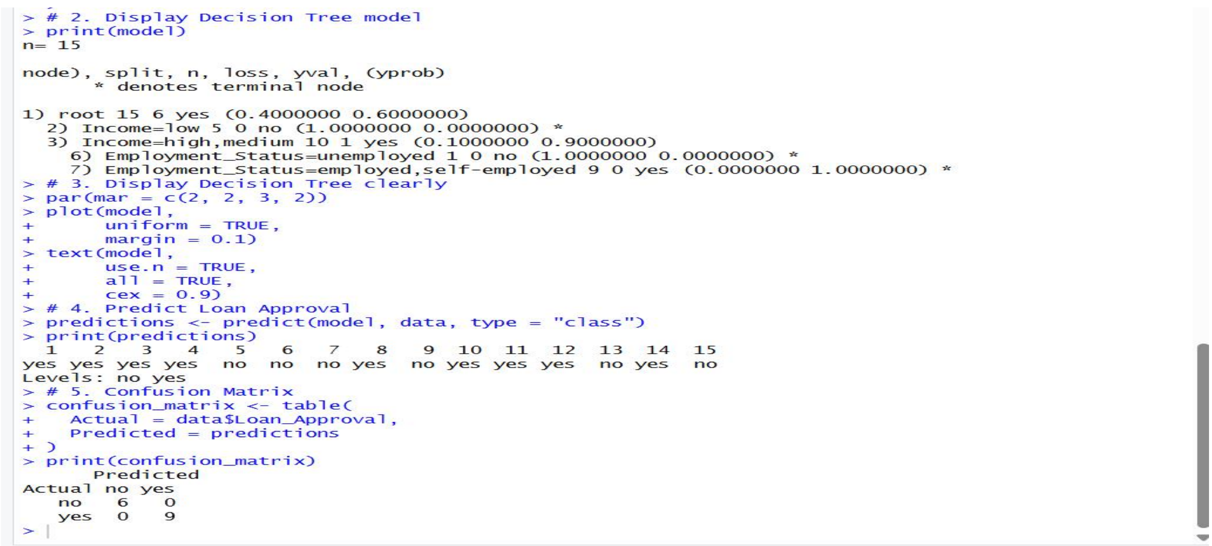


```
> # Enter data
> data <- data.frame(
+   Age = c("young", "young", "middle", "old", "old",
+           "old", "middle", "young", "young", "old",
+           "middle", "middle", "old", "young", "middle"),
+   Income = c("high", "high", "high", "medium", "low",
+              "low", "low", "medium", "low", "medium",
+              "medium", "high", "medium", "medium", "low"),
+   Employment_Status = c("employed", "self-employed", "employed",
+                         "employed", "unemployed", "self-employed",
+                         "unemployed", "employed", "unemployed",
+                         "self-employed", "employed", "self-employed",
+                         "unemployed", "self-employed", "employed"),
+   Credit_Score = c("good", "average", "good", "good", "poor",
+                    "average", "poor", "average", "poor", "good",
+                    "average", "good", "poor", "average", "poor"),
+   Loan_Approval = c("yes", "yes", "yes", "yes", "no",
+                     "no", "no", "yes", "no", "yes",
+                     "yes", "yes", "no", "yes", "no")
+ )
> # Display data
> print(data)
      Age Income Employment_Status Credit_Score Loan_Approval
1   young   high          employed         good           yes
2   young   high     self-employed      average           yes
3  middle   high          employed         good           yes
4     old medium          employed         good           yes
5     old    low        unemployed         poor            no
6     old    low     self-employed      average            no
7  middle    low        unemployed         poor            no
8   young medium          employed      average           yes
9   young    low        unemployed         poor            no
10    old medium     self-employed         good           yes
11 middle medium          employed      average           yes
12 middle   high     self-employed         good           yes
13    old medium        unemployed         poor            no
14  young medium     self-employed      average           yes
15 middle    low          employed         poor            no
> # Convert variables to factors
> data$Age <- as.factor(data$Age)
```

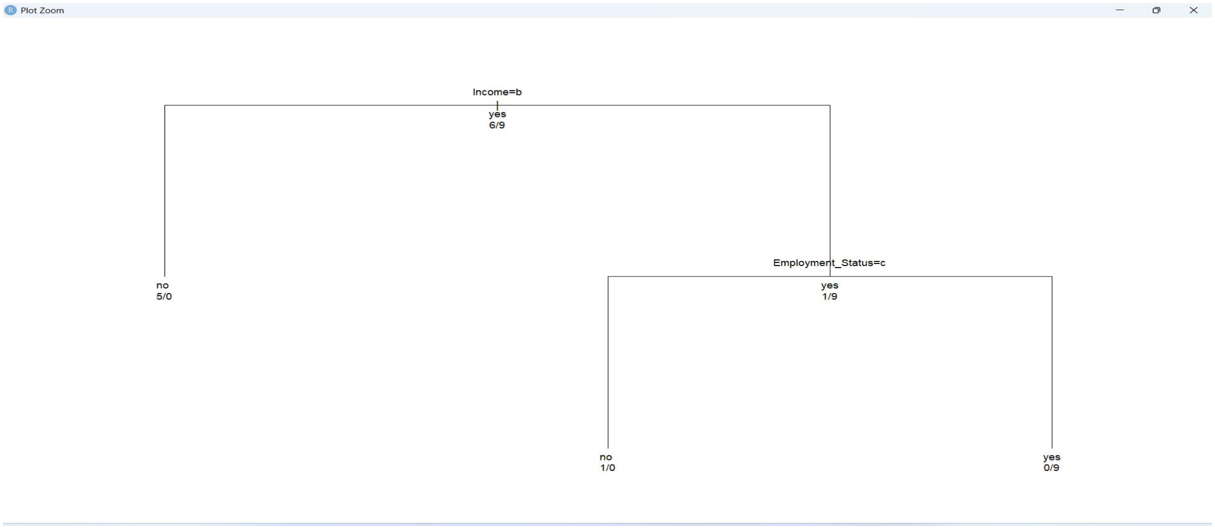
```
Data
data                15 obs. of 5 variables
df                  3 obs. of 3 variables
model               List of  15
Values
age                 num [1:3] 23 24 25
confusion_matrix    'table' int [1:2, 1:2] 6 0 0 9
marks               num [1:3] 88 78 25
names               chr [1:3] "siri" "mahi" "chiru"
predictions         Factor w/ 2 levels "no","yes": 2 2 2 2 1 1 1 2 1 2 ...
```

```
> # Convert variables to factors
> data$Age <- as.factor(data$Age)
> data$Income <- as.factor(data$Income)
> data$Employment_Status <- as.factor(data$Employment_Status)
> data$Credit_Score <- as.factor(data$Credit_Score)
> data$Loan_Approval <- as.factor(data$Loan_Approval)
> # 1. Build Decision Tree classifier
> library(rpart)
> model <- rpart(
+   Loan_Approval ~ Age + Income + Employment_Status + Credit_Score,
+   data = data,
+   method = "class",
+   control = rpart.control(
+     cp = 0,
+     minsplit = 2,
+     minbucket = 1,
+     maxdepth = 5
+   )
+ )
> # 2. Display Decision Tree model
> print(model)
n= 15

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 15 6 yes (0.4000000 0.6000000)
  2) Income=low 5 0 no (1.0000000 0.0000000) *
  3) Income=high,medium 10 1 yes (0.1000000 0.9000000)
    6) Employment_Status=unemployed 1 0 no (1.0000000 0.0000000) *
    7) Employment_Status=employed,self-employed 9 0 yes (0.0000000 1.0000000) *
> # 3. Display Decision Tree clearly
> par(mar = c(2, 2, 3, 2))
> plot(model,
+      uniform = TRUE,
+      margin = 0.1)
> text(model,
+      use.n = TRUE,
+      all = TRUE,
+      cex = 0.9)
```



```
> # 2. Display Decision Tree model
> print(model)
n= 15

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 15 6 yes (0.4000000 0.6000000)
  2) Income=low 5 0 no (1.0000000 0.0000000) *
  3) Income=high,medium 10 1 yes (0.1000000 0.9000000)
    6) Employment_Status=unemployed 1 0 no (1.0000000 0.0000000) *
    7) Employment_Status=employed,self-employed 9 0 yes (0.0000000 1.0000000) *
> # 3. Display Decision Tree clearly
> par(mar = c(2, 2, 3, 2))
> plot(model,
+      uniform = TRUE,
+      margin = 0.1)
> text(model,
+      use.n = TRUE,
+      all = TRUE,
+      cex = 0.9)
> # 4. Predict Loan Approval
> predictions <- predict(model, data, type = "class")
> print(predictions)
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15
yes yes yes yes  no  no  no yes  no yes yes yes  no yes  no
Levels: no yes
> # 5. Confusion Matrix
> confusion_matrix <- table(
+   Actual = data$Loan_Approval,
+   Predicted = predictions
+ )
> print(confusion_matrix)
      Predicted
Actual no yes
   no   6   0
   yes  0   9
>
```



4. Results and Interpretation

- The decision tree predicts **Loan Approval** mainly using Income and Employment Status. Low-income applicants are predicted as **No**, while high/medium-income applicants are further classified based on Employment Status.
- The confusion matrix shows **6 correctly predicted No cases and 9 correctly predicted Yes cases**. Therefore, all 15 observations were correctly classified.

5. Model Performance

- The model accuracy is:
- **Accuracy = 15/15 × 100 = 100%**
- There are no incorrect predictions in the given dataset, indicating that the model has classified all observations correctly.

6. Limitations of the Study

The dataset contains only **15 observations**, which is relatively small for evaluating a classification model. The model was also evaluated using the same data used for training. Therefore, the 100% accuracy may not represent how well the model performs on new or unseen applicants. A larger dataset and separate testing data would provide a more reliable evaluation of the model.

7. Conclusion

- The decision tree successfully predicts loan approval for the given dataset with **100% classification accuracy**. The results show that **Income and Employment Status** are the main factors used by the model to determine loan approval.
- However, because the dataset is small, the results should be interpreted carefully. Further analysis using a larger dataset would help determine whether the model can provide reliable predictions for new loan applicants.

QUESTION 04: Year 9 Mathematics Class Performance

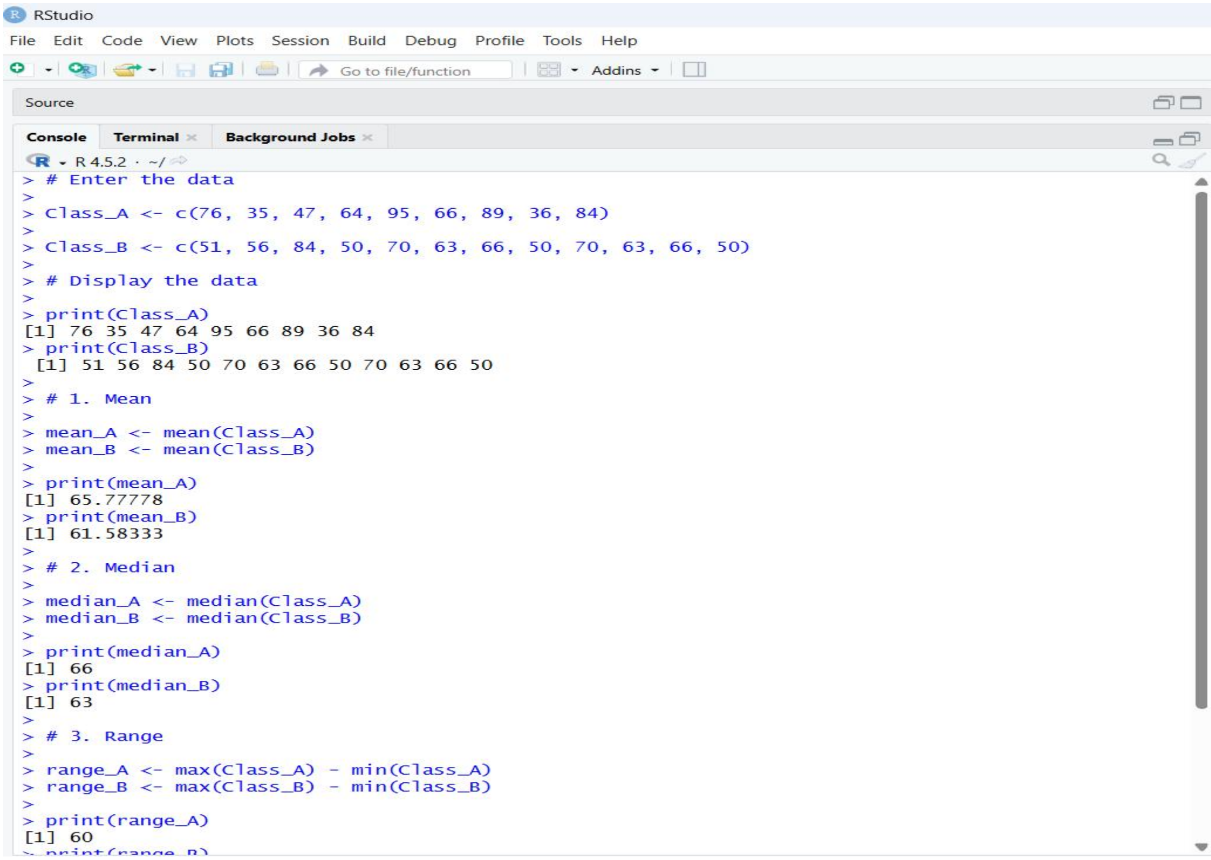
1. Introduction

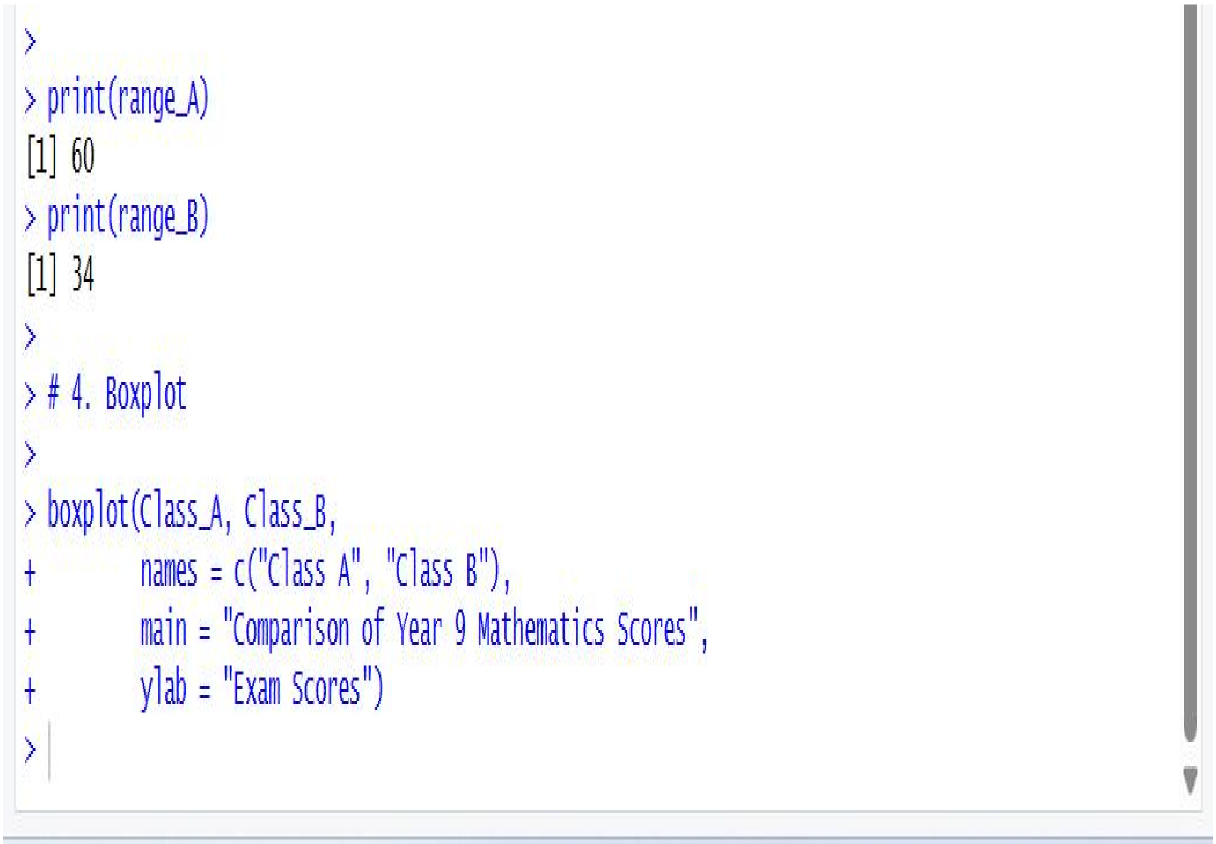
This question compares the examination performance of two Year 9 mathematics classes. The mean, median and range are used to compare the overall performance and spread of scores. A boxplot is also used to visually compare the distribution of scores and identify any possible differences or unusual values between the two classes.

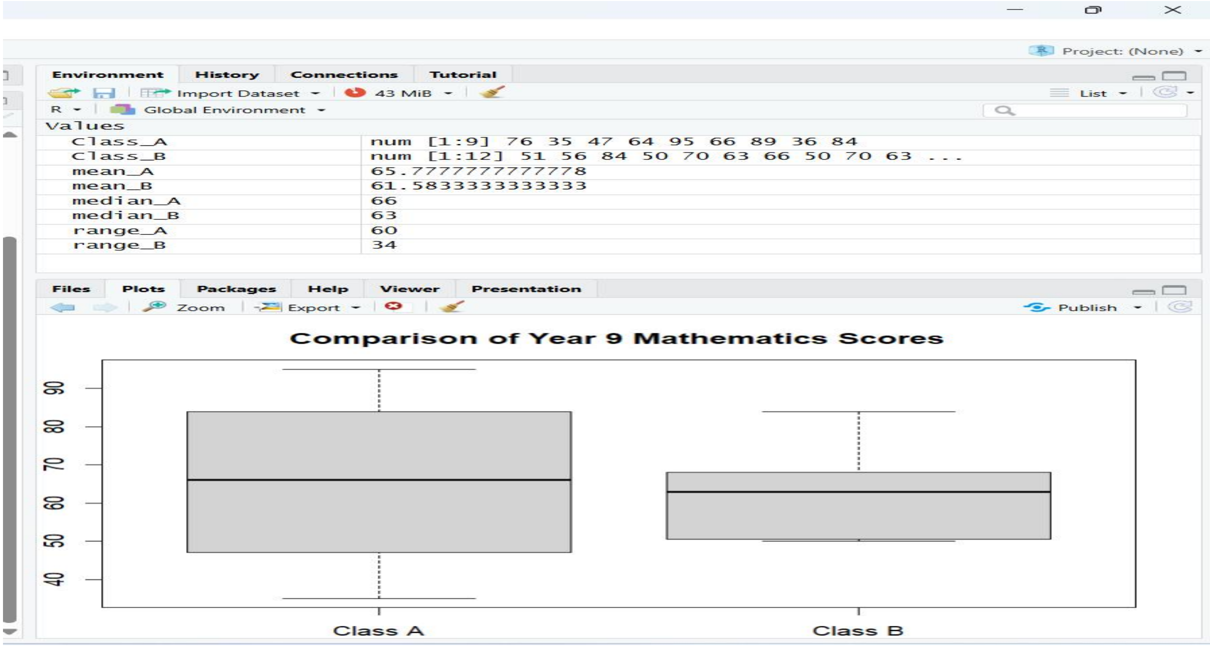
2. R Code

```
# Enter the data
Class_A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
Class_B <- c(51, 56, 84, 50, 70, 63, 66, 50, 70, 63, 66, 50)


# Display the data
print(Class_A)print(Class_B)


# 1. Mean
mean_A <- mean(Class_A)mean_B <- mean(Class_B)
print(mean_A)print(mean_B)


# 2. Median
median_A <- median(Class_A)median_B <- median(Class_B)
print(median_A)print(median_B)


# 3. Range
range_A <- max(Class_A) - min(Class_A)range_B <- max(Class_B) - min(Class_B)
print(range_A)print(range_B)
# 4. Boxplot
boxplot(Class_A, Class_B,
    names = c("Class A", "Class B"),
    main = "Comparison of Year 9 Mathematics Scores",
    ylab = "Exam Scores")
```

3. OUTPUT


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Source
Console
Terminal
Background Jobs
R 4.5.2 · ~/
> # Enter the data
>
> Class_A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
>
> Class_B <- c(51, 56, 84, 50, 70, 63, 66, 50, 70, 63, 66, 50)
>
> # Display the data
>
> print(Class_A)
[1] 76 35 47 64 95 66 89 36 84
> print(Class_B)
[1] 51 56 84 50 70 63 66 50 70 63 66 50
>
> # 1. Mean
>
> mean_A <- mean(Class_A)
> mean_B <- mean(Class_B)
>
> print(mean_A)
[1] 65.77778
> print(mean_B)
[1] 61.58333
>
> # 2. Median
>
> median_A <- median(Class_A)
> median_B <- median(Class_B)
>
> print(median_A)
[1] 66
> print(median_B)
[1] 63
>
> # 3. Range
>
> range_A <- max(Class_A) - min(Class_A)
> range_B <- max(Class_B) - min(Class_B)
>
> print(range_A)
[1] 60


>
> print(range_A)
[1] 60
> print(range_B)
[1] 34
>
> # 4. Boxplot
>
> boxplot(Class_A, Class_B,
+ names = c("Class A", "Class B"),
+ main = "Comparison of Year 9 Mathematics Scores",
+ ylab = "Exam Scores")
>



4. Results and Interpretation

- The calculated results show that **Class A performed slightly better overall** than Class B. Class A has a higher mean score of **65.78** compared with **61.58** for Class B. The median score is also higher for Class A (**66**) than for Class B (**63**).
- From the boxplot, Class A has a wider spread of scores, while Class B's scores are more closely concentrated around the middle. This indicates that Class A had both higher typical performance and greater variation in students' scores.

5. Outlier Analysis

- The boxplot does **not show any individual outlier points** for either Class A or Class B. Therefore, there is no clear evidence of an unusually high or low score that needs to be treated separately.
- Class A has a wider overall spread, with scores ranging from **35 to 95**, whereas Class B ranges from **50 to 84**. Hence, Class A shows greater variation in performance.

6. Conclusion

- Based on the mean, median and boxplot, **Class A performed better overall** than Class B. Class A has the higher mean (**65.78**) and median (**66**), compared with a mean of **61.58** and median of **63** for Class B.
- However, Class A has a larger range (**60**) than Class B (**34**), showing that the performance of students in Class A was more variable. Class B's scores were more consistent, although its overall performance was slightly lower.
- Therefore, **Class A achieved better overall results, while Class B showed more consistent performance**.

QUESTION 05 – Age and Body Fat Data

1. Introduction

This question analyzes the **age and body fat percentage (%fat)** of 18 randomly selected adults. The analysis will summarize both variables using measures of central tendency and variation, and will also examine their distributions and relationship.

The required analysis includes calculating the **mean, median, and standard deviation** for age and %fat, drawing **boxplots** for both variables, and creating a **scatter plot and Q-Q plot** to study the relationship and distribution of the data.

2. R Code

 # Enter the data

age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50,

    52, 54, 54, 56, 57, 58, 58, 60, 61)

fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,

    34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)

 # Display the data

print(age)

print(fat)

 # (a) Mean, Median and Standard Deviation

Age

mean_age <- mean(age)

median_age <- median(age)

sd_age <- sd(age)

Body Fat

mean_fat <- mean(fat)

median_fat <- median(fat)

```
sd_fat <- sd(fat)

# Display results

print(mean_age)

print(median_age)

print(sd_age)

print(mean_fat)

print(median_fat)

print(sd_fat)

# (b) Boxplots

# Boxplot for Age

boxplot(age,

        main = "Boxplot of Age",

        ylab = "Age")

# Boxplot for Body Fat

boxplot(fat,

        main = "Boxplot of Body Fat Percentage",

        ylab = "% Body Fat")

# (c) Scatter Plot

plot(age, fat,

     main = "Scatter Plot of Age vs Body Fat",

     xlab = "Age",

     ylab = "% Body Fat",

     pch = 19)

# Add a regression line
```

```
abline(lm(fat ~ age))
```

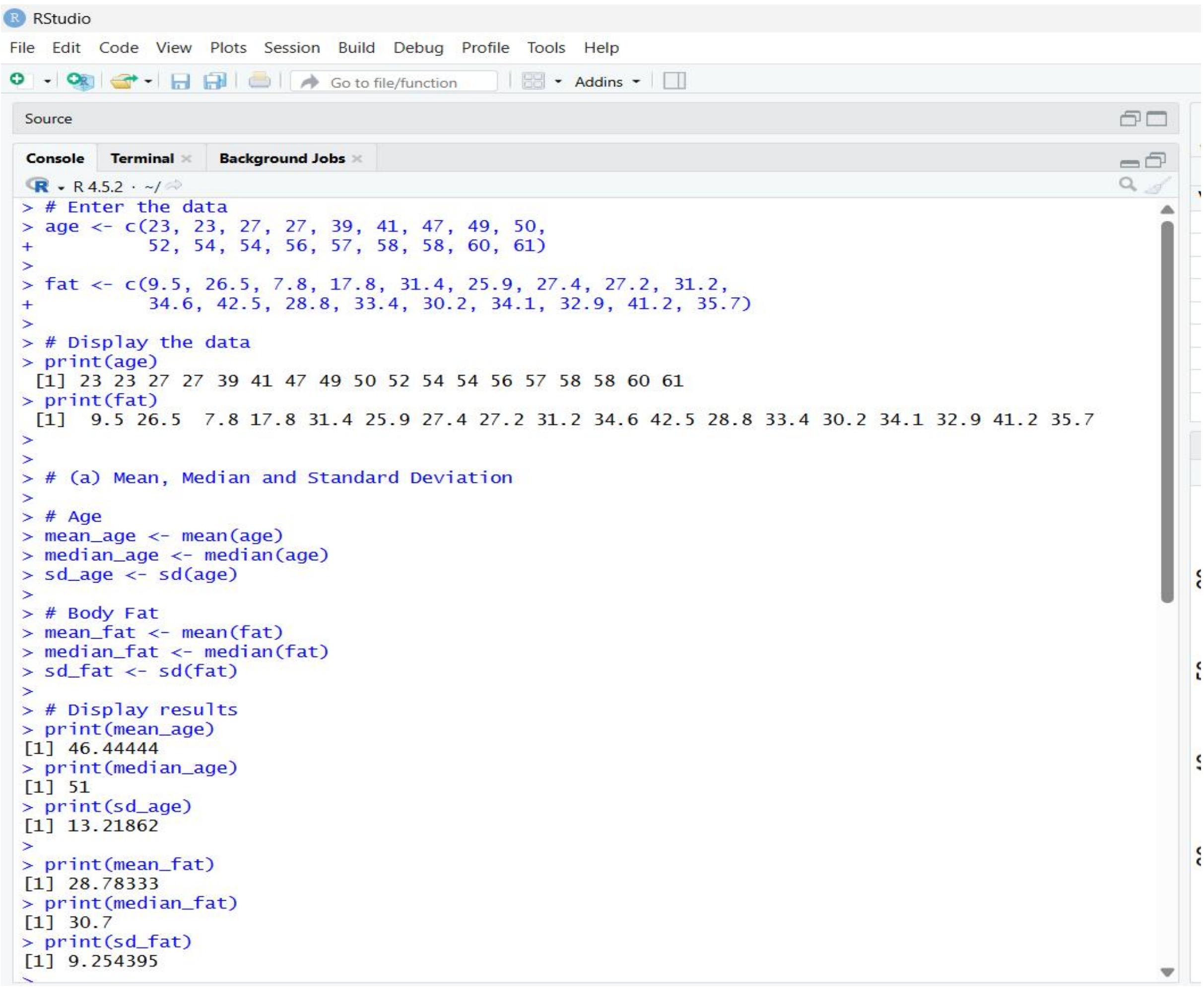
 # Q-Q Plot for Age

```
qqnorm(age,
       main = "Q-Q Plot of Age")
qqline(age)
```

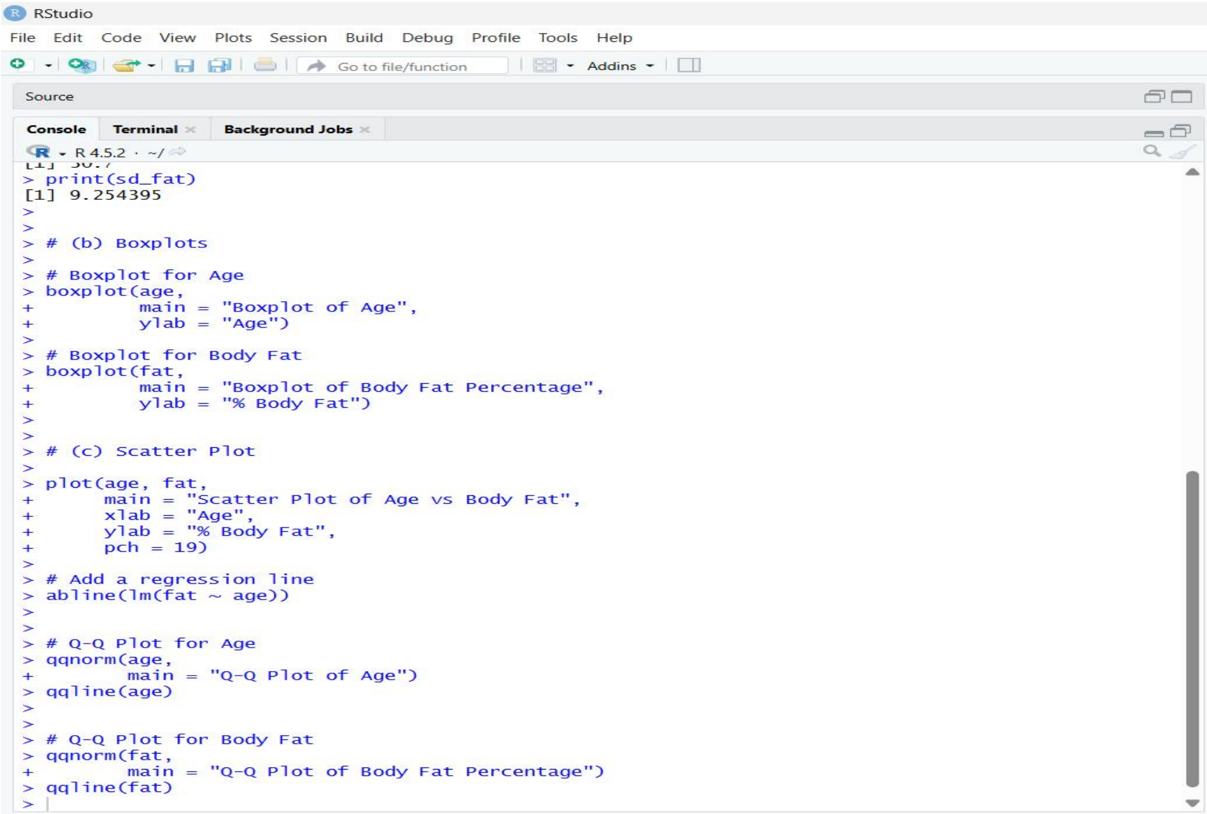
 # Q-Q Plot for Body Fat

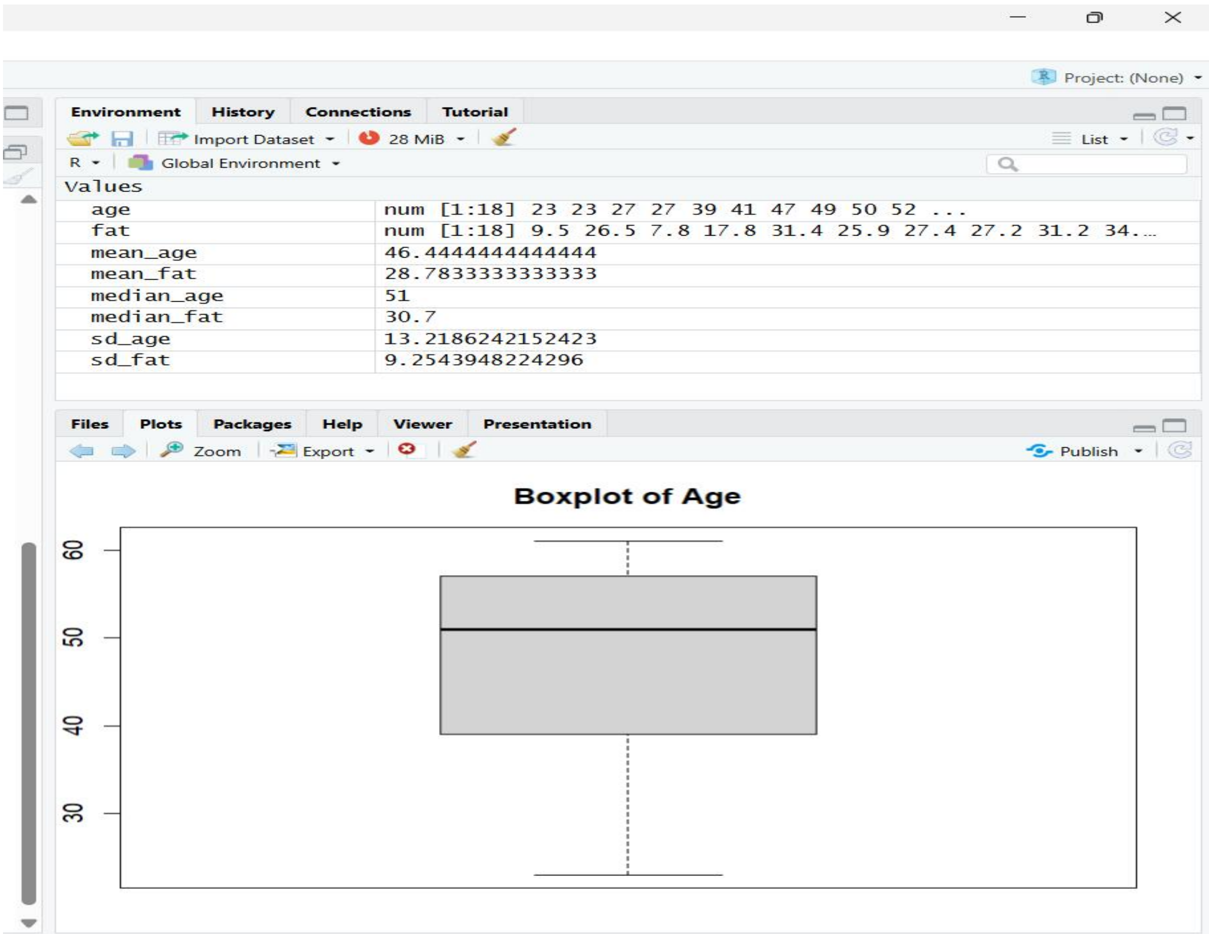
```
qqnorm(fat,
       main = "Q-Q Plot of Body Fat Percentage")
qqline(fat)
```

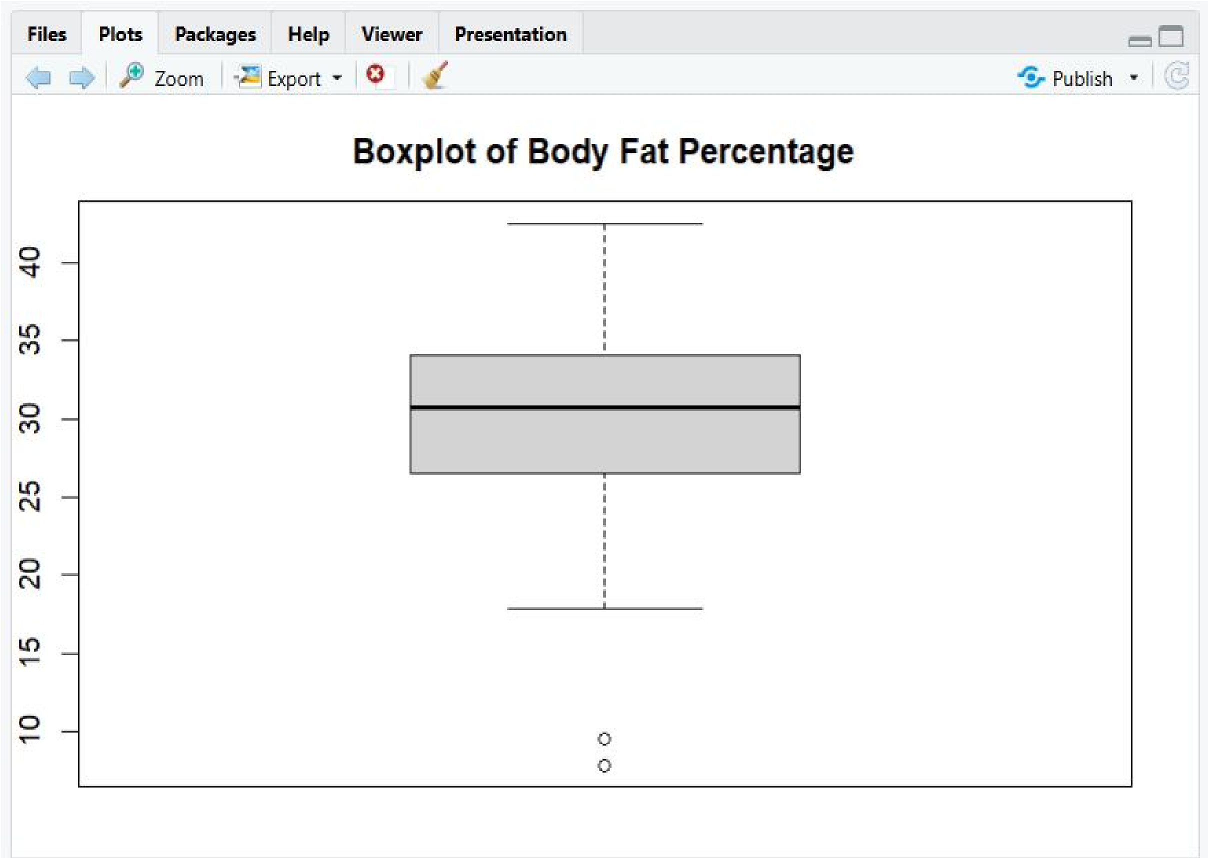
3. OUTPUT



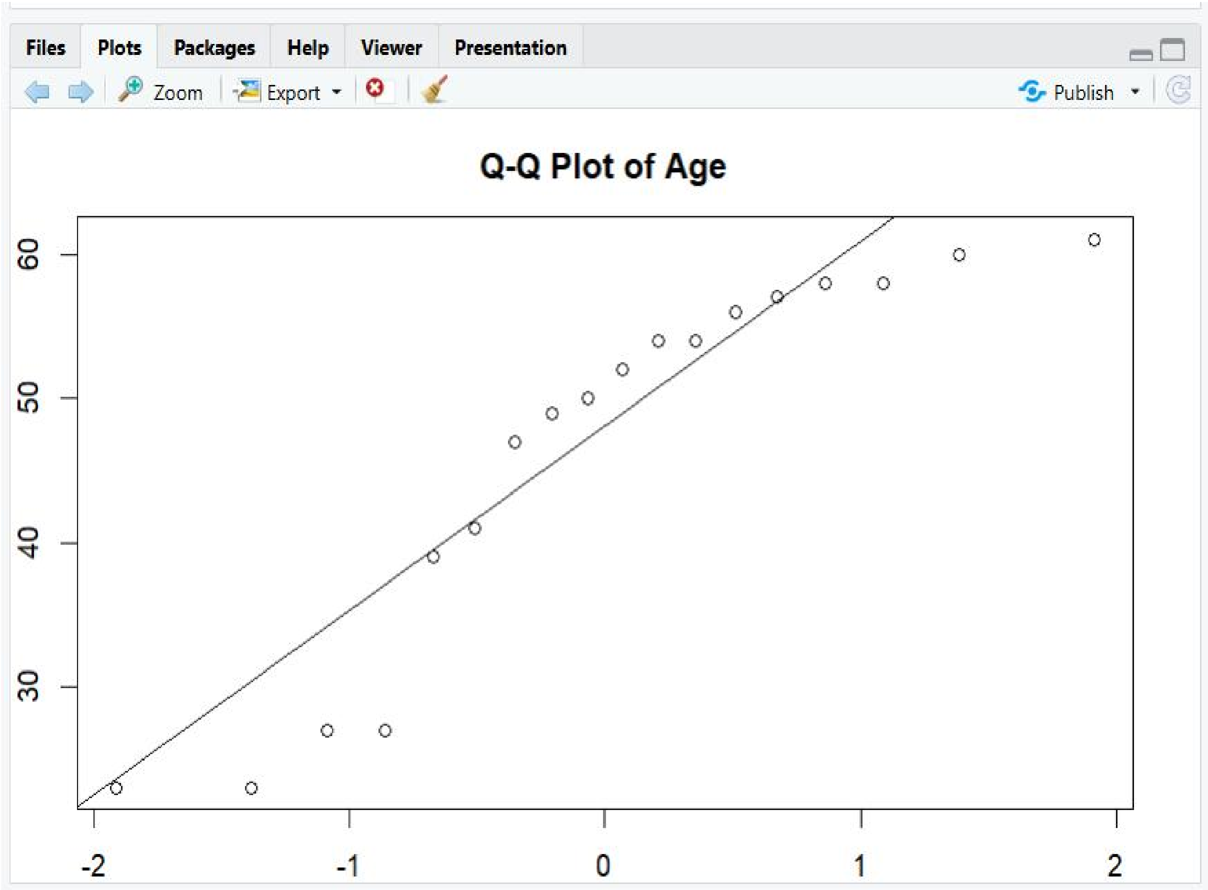

```
> # Enter the data
> age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50,
+          52, 54, 54, 56, 57, 58, 58, 60, 61)
>
> fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
+          34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)
>
> # Display the data
> print(age)
 [1] 23 23 27 27 39 41 47 49 50 52 54 54 56 57 58 58 60 61
> print(fat)
 [1]  9.5 26.5  7.8 17.8 31.4 25.9 27.4 27.2 31.2 34.6 42.5 28.8 33.4 30.2 34.1 32.9 41.2 35.7
>
>
> # (a) Mean, Median and Standard Deviation
>
> # Age
> mean_age <- mean(age)
> median_age <- median(age)
> sd_age <- sd(age)
>
> # Body Fat
> mean_fat <- mean(fat)
> median_fat <- median(fat)
> sd_fat <- sd(fat)
>
> # Display results
> print(mean_age)
[1] 46.44444
> print(median_age)
[1] 51
> print(sd_age)
[1] 13.21862
>
> print(mean_fat)
[1] 28.78333
> print(median_fat)
[1] 30.7
> print(sd_fat)
[1] 9.254395
>
```

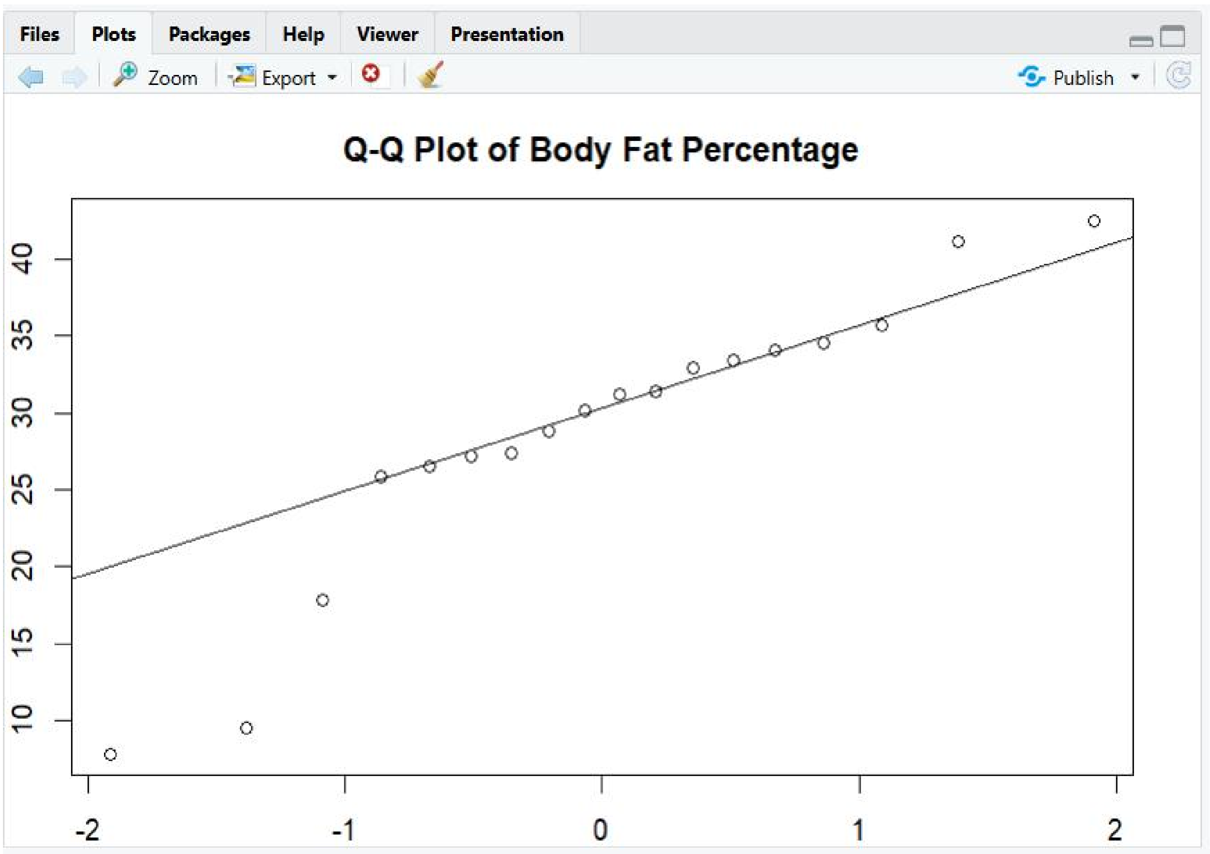

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Source
Console
Terminal
Background Jobs
R 4.5.2 · ~/
> print(sd_fat)
[1] 9.254395
>
>
> # (b) Boxplots
>
> # Boxplot for Age
> boxplot(age,
+ main = "Boxplot of Age",
+ ylab = "Age")
>
> # Boxplot for Body Fat
> boxplot(fat,
+ main = "Boxplot of Body Fat Percentage",
+ ylab = "% Body Fat")
>
>
> # (c) Scatter Plot
>
> plot(age, fat,
+ main = "Scatter Plot of Age vs Body Fat",
+ xlab = "Age",
+ ylab = "% Body Fat",
+ pch = 19)
>
> # Add a regression line
> abline(lm(fat ~ age))
>
>
> # Q-Q Plot for Age
> qqnorm(age,
+ main = "Q-Q Plot of Age")
> qqline(age)
>
>
> # Q-Q Plot for Body Fat
> qqnorm(fat,
+ main = "Q-Q Plot of Body Fat Percentage")
> qqline(fat)
>


Project: (None)
Environment
History
Connections
Tutorial
Import Dataset
28 MiB
List
R
Global Environment
Values
age num [1:18] 23 23 27 27 39 41 47 49 50 52 ...
fat num [1:18] 9.5 26.5 7.8 17.8 31.4 25.9 27.4 27.2 31.2 34....
mean_age 46.4444444444444
mean_fat 28.7833333333333
median_age 51
median_fat 30.7
sd_age 13.2186242152423
sd_fat 9.2543948224296
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Boxplot of Age
30
40
50
60


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Boxplot of Body Fat Percentage
10
15
20
25
30
35
40


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Scatter Plot of Age vs Body Fat
10
15
20
25
30
35
40
30
40
50
60


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Q-Q Plot of Age
30
40
50
60
-2
-1
0
1
2


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Q-Q Plot of Body Fat Percentage
10
15
20
25
30
35
40
-2
-1
0
1
2

4. Results and Interpretation

(a) Mean, Median and Standard Deviation

Variable	Mean	Median	SD
Age	46.44	51	13.22
Body Fat (%)	28.78	30.7	9.25

- The average age is **46.44 years**, while the average body fat is **28.78%**. Age has greater variation than body fat percentage because its standard deviation is higher.

(b) Boxplots

The boxplots show the spread and distribution of **age** and **body fat percentage**. Age has a wider spread, while body fat percentage is comparatively less variable.

(c) Scatter Plot and Q-Q Plot

The scatter plot shows a **positive relationship** between age and body fat percentage. The Q-Q plots show that most observations approximately follow the normal pattern, with some deviations at the ends.

5. Conclusion

The results show that **age and body fat percentage have a positive relationship** in this sample. As age increases, body fat generally tends to increase. Age also shows greater variability than body fat percentage.

Limitation

The sample contains only **18 adults**, so the results may not represent the wider population. Also, the relationship observed does not prove that age alone causes changes in body fat percentage.